

EDR vs Antivirus: The Modern Endpoint

Why signatures lost

AV hashes a file; evaders re-pack. EDR watches **what endpoints do continuously** and relates process/net/registry behavior in time – because behavior survives repacking.

The EDR model

```
[ Agent ]
| collects: process (create/kill), file (open/write), registry,
|           network (connections/ports), DLL loads, scripts, wmi
v
[ Cloud console ] can correlate across endpoints (hunting, rules)

Alert pipeline:
endpoint -> stream -> detection engines -> analyst(s) -> response action
```

Capabilities that matter

Process genealogy (parent→child): 'Word spawned PowerShell' = lead

Behavioral rules: powershell download cradle, LSASS read, wmi spawning

Live response: kill, quarantine, terminate process, take memory/image

Hunting UI: search across events (my new file, evtX)

Rollback: some EDRs offer 'restore point' after ransomware

XDR = the expanded view

EDR + network, cloud, email, identity telemetry into one platform – detections span endpoints *and* the wire *and* the org's services.

Trade-offs

Cost, tuning burden, privacy (endpoint telemetry is sensitive)

EDR is not a silver bullet: attacker TTPs still escape; analysts matter

Alert flooding is the #1 EDR failure mode – tuning rules + baselines is the real job

Detection-first posture

The EDR is only as good as the rules on it: process trees, LSASS protection, macro/script usage, anomalous admin usage, cred-dump signs, persistence writes.

EDR shifts the battlefield from files to behavior. Your job as a future analyst/hunter/engineer is to make the behavior rules sharp, the baselines real, and the response fast.

[← Module start](#)

[Telemetry Sources: OS Events, ...](#)